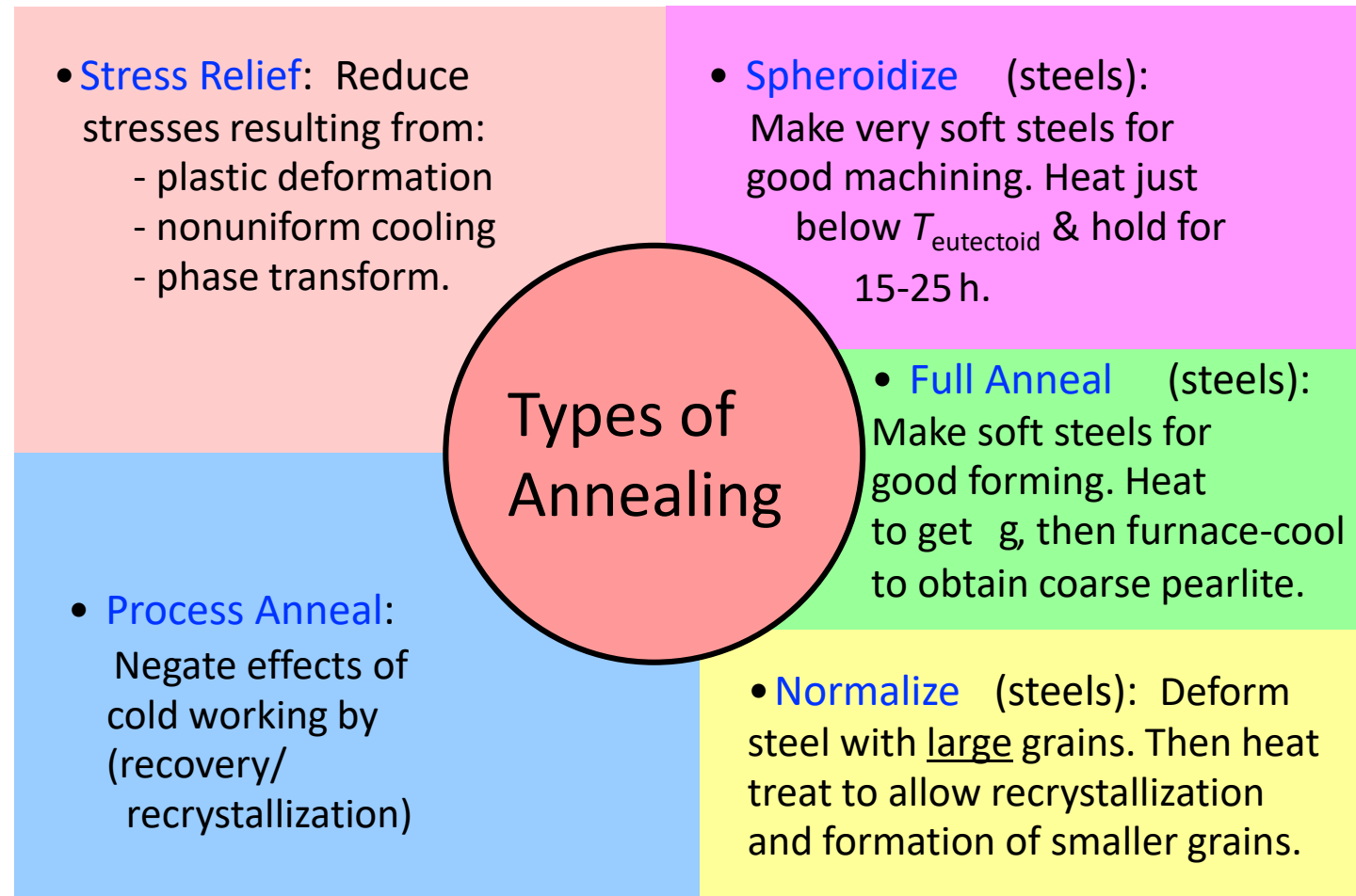


Processing of Metals 2

Presented by Dr Adijat 'Wumi Inyang

Thermal Processing of Metals

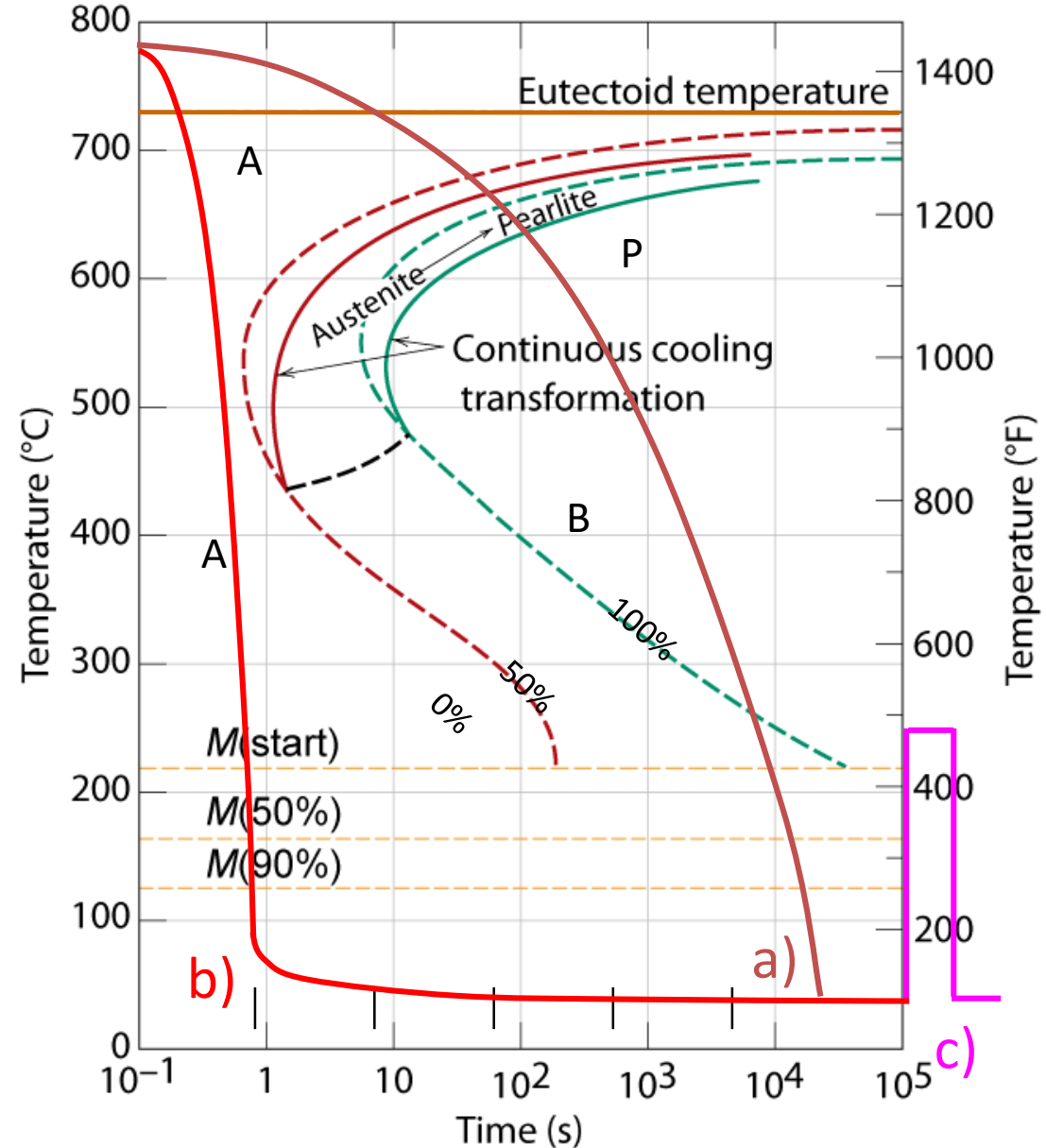
Annealing: Heat to T_{anneal} , then cool slowly.



Heat Treatment Temperature-Time Paths

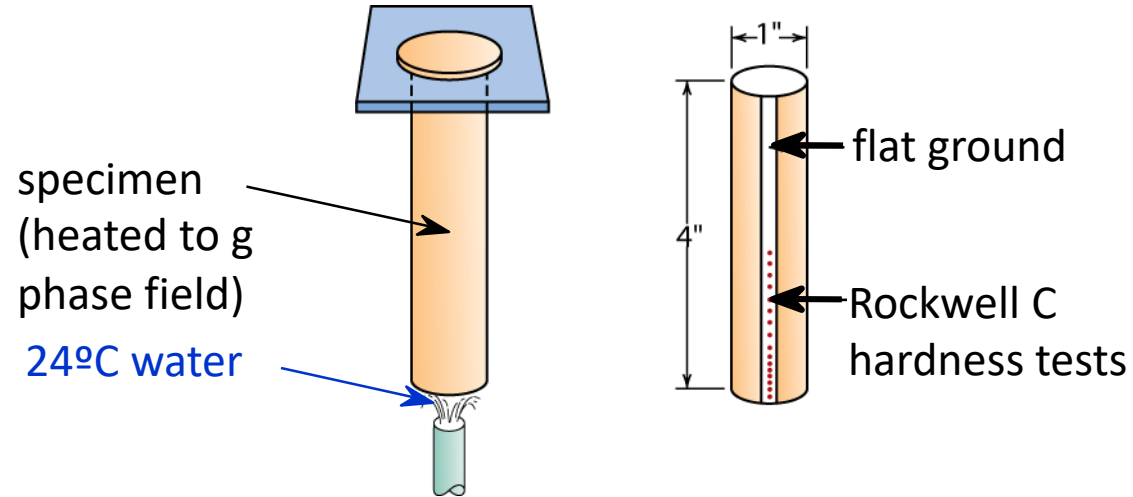
- a) Full Annealing
- b) Quenching
- c) Tempering (Tempered Martensite)

Fig. 10.25,
Callister &
Rethwisch 8e.



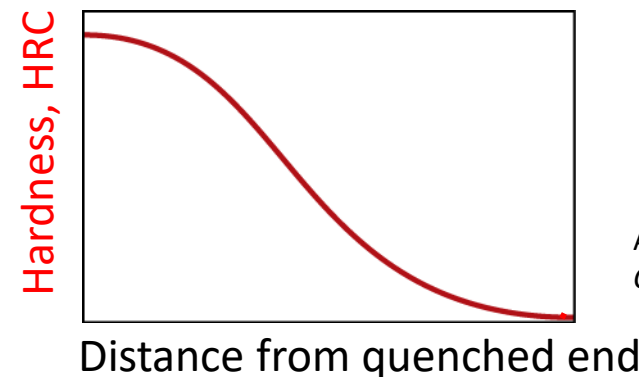
Hardenability -- Steels

- Hardenability – measure of the ability to form martensite
- Jominy end quench test used to measure hardenability.



Adapted from Fig. 11.11,
Callister & Rethwisch 8e. (Fig.
11.11 adapted from A.G. Guy,
*Essentials of Materials
Science*, McGraw-Hill Book
Company, New York, 1978.)

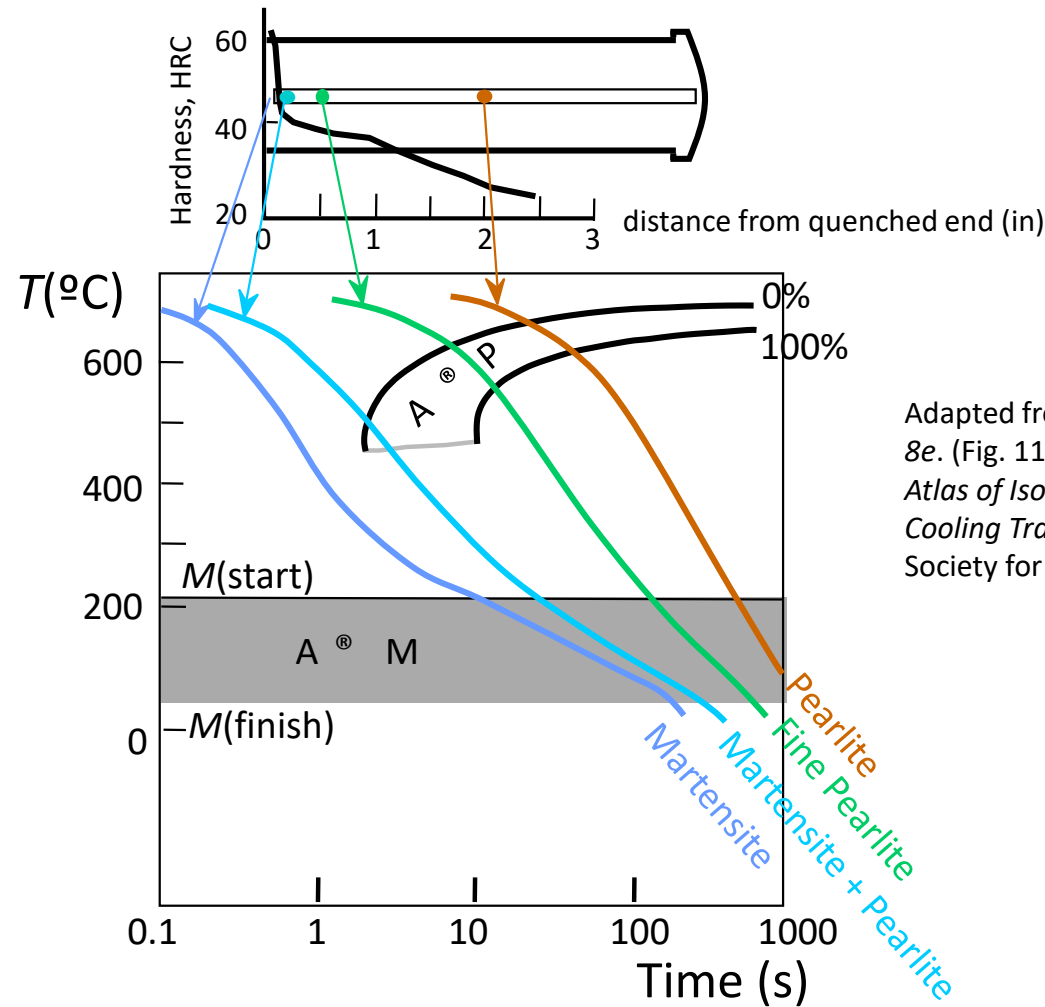
- Plot hardness versus distance from the quenched end.



Adapted from Fig. 11.12,
Callister & Rethwisch 8e.

Reason Why Hardness Changes with Distance

- The cooling rate decreases with distance from quenched end.

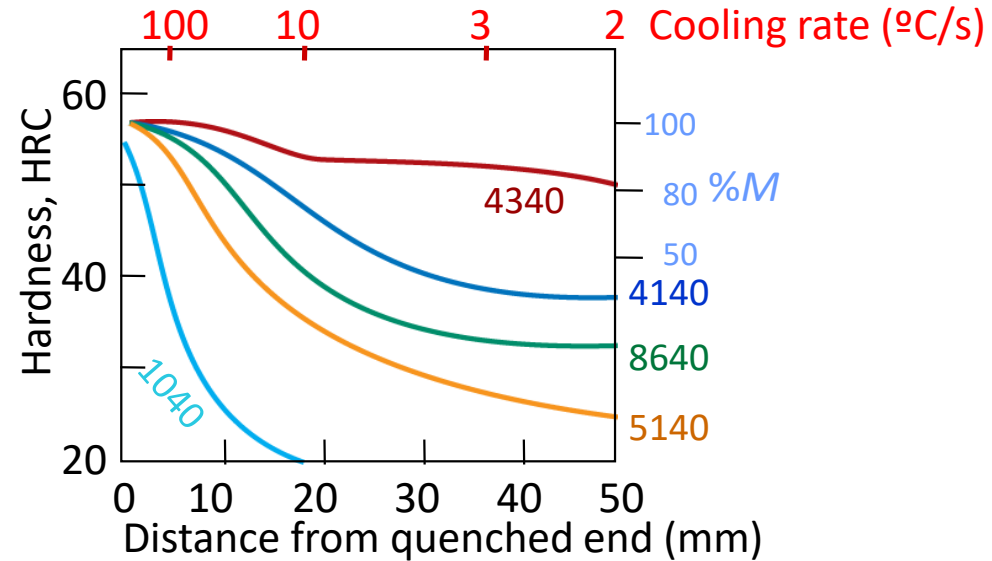


Adapted from Fig. 11.13, *Callister & Rethwisch 8e.* (Fig. 11.13 adapted from H. Boyer (Ed.) *Atlas of Isothermal Transformation and Cooling Transformation Diagrams*, American Society for Metals, 1977, p. 376.)

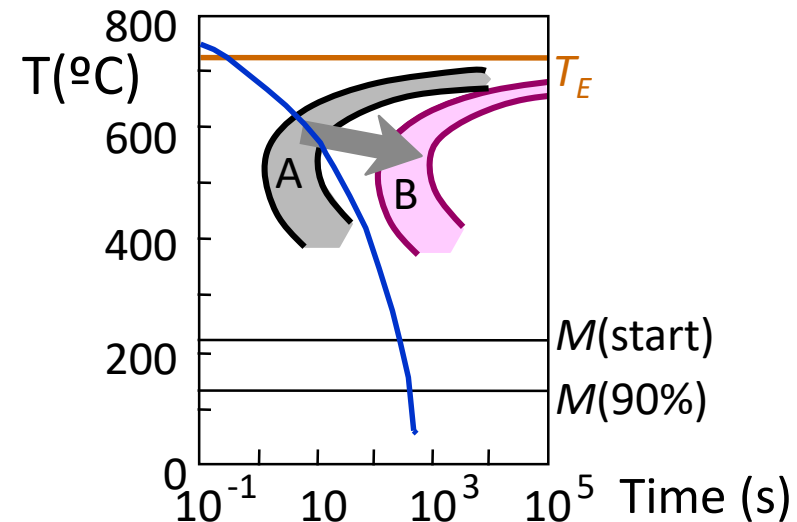
Hardenability vs Alloy Composition

- Hardenability curves for five alloys each with, $C = 0.4 \text{ wt\% C}$

Adapted from Fig. 11.14, *Callister & Rethwisch 8e*. (Fig. 11.14 adapted from figure furnished courtesy Republic Steel Corporation.)



- "Alloy Steels"
(4140, 4340, 5140, 8640)
 - contain Ni, Cr, Mo (0.2 to 2 wt%)
 - these elements shift the "nose" to longer times (from A to B)
 - martensite is easier to form



Influences of Quenching Medium & Specimen Geometry

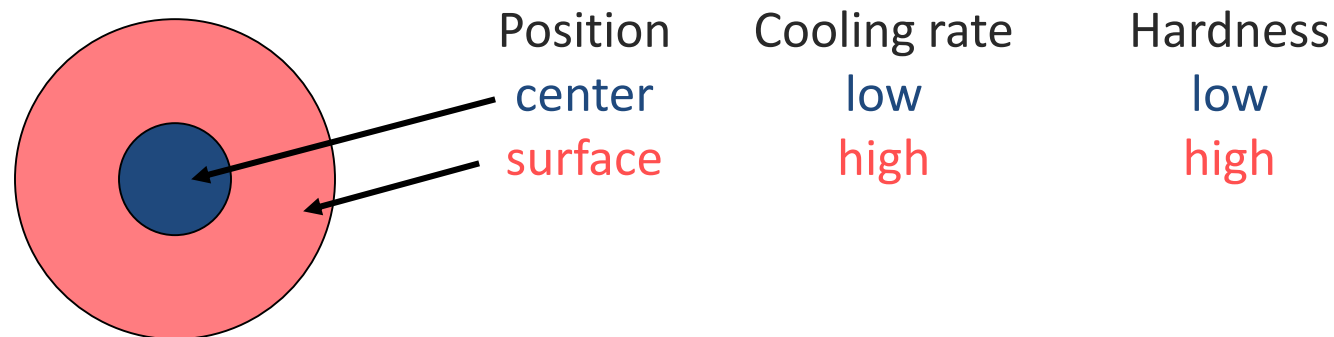
- Effect of quenching medium:

Medium	Severity of Quench	Hardness
air	low	low
oil	moderate	moderate
water	high	high

- Effect of specimen geometry:

When surface area-to-volume ratio increases:

- cooling rate throughout interior increases
- hardness throughout interior increases



Precipitation Hardening

- Particles impede dislocation motion.

- Ex: Al-Cu system

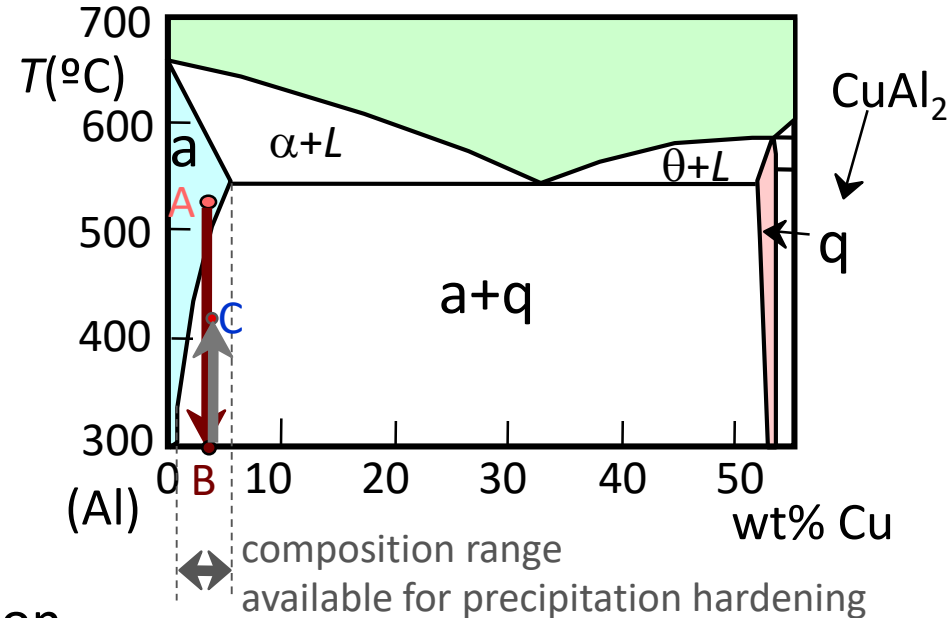
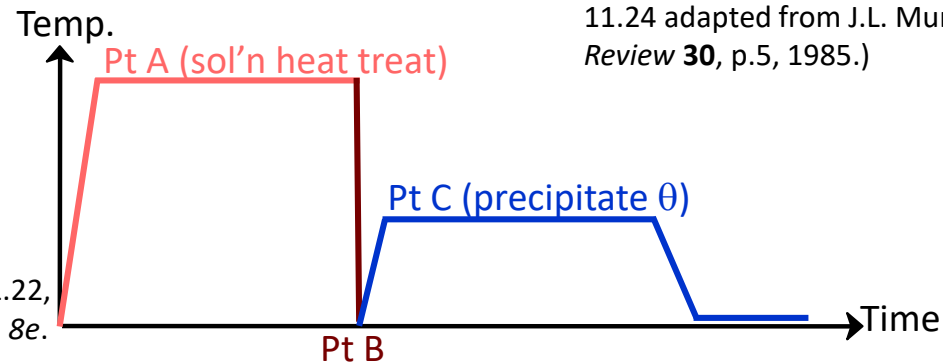
- Procedure:

- Pt A: solution heat treat (get a solid solution)
- Pt B: quench to room temp. (retain a solid solution)
- Pt C: reheat to nucleate small q particles within a phase.

- Other alloys that precipitation harden:

- Cu-Be
- Cu-Sn
- Mg-Al

Adapted from Fig. 11.22,
Callister & Rethwisch 8e.



Adapted from Fig. 11.24, Callister & Rethwisch 8e. (Fig. 11.24 adapted from J.L. Murray, *International Metals Review* 30, p.5, 1985.)

Application: Precipitation Strengthening

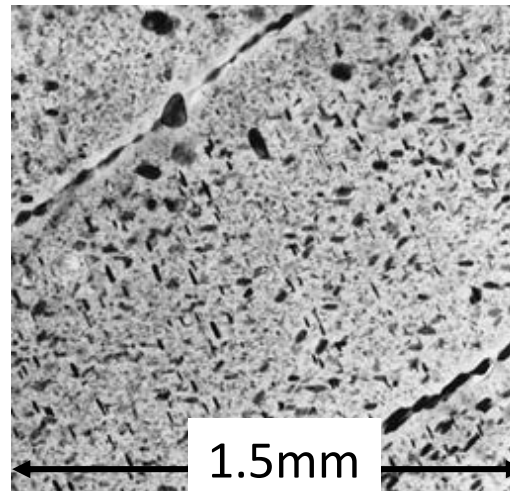


- Internal wing structure on Boeing 767



Adapted from chapter-opening photograph, Chapter 11, *Callister & Rethwisch 3e*. (courtesy of G.H. Narayanan and A.G. Miller, Boeing Commercial Airplane Company.)

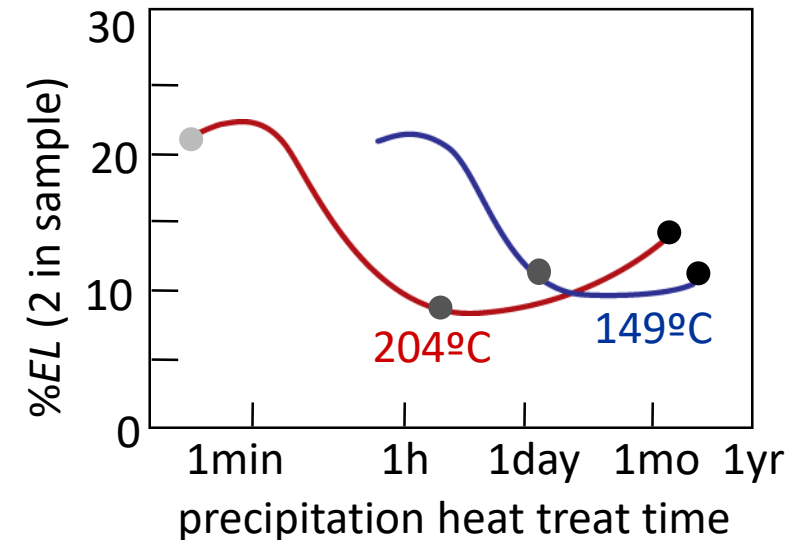
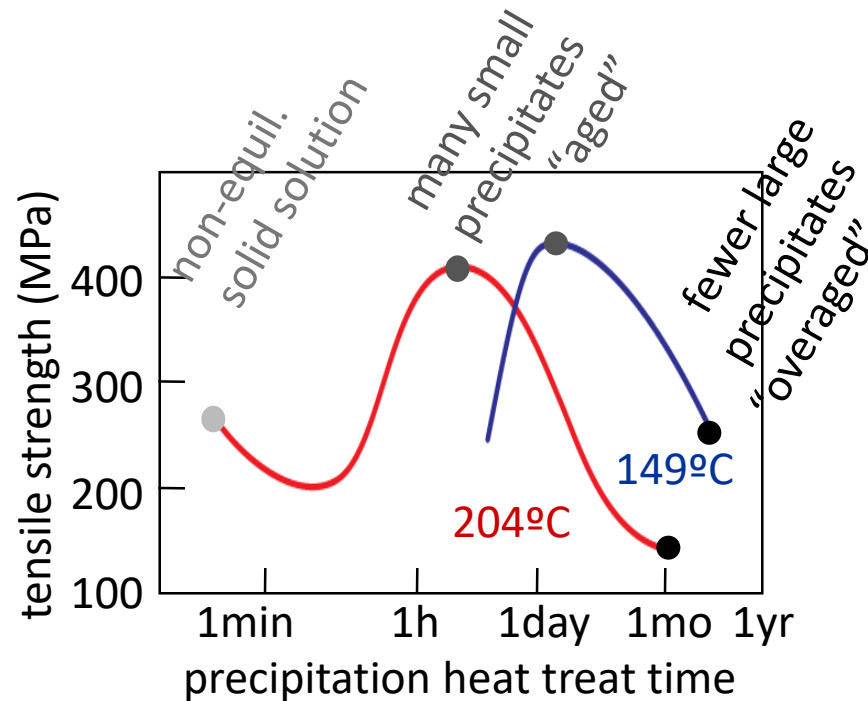
- Aluminum is strengthened with precipitates formed by alloying.



Adapted from Fig. 11.26, *Callister & Rethwisch 8e*. (Fig. 11.26 is courtesy of G.H. Narayanan and A.G. Miller, Boeing Commercial Airplane Company.)

Influence of Precipitation Heat Treatment on TS , % EL

- 2014 Al Alloy:
- Maxima on TS curves.
- Increasing T accelerates process.
- Minima on % EL curves.



Adapted from Fig. 11.27, Callister & Rethwisch 8e. (Fig. 11.27 adapted from *Metals Handbook: Properties and Selection: Nonferrous Alloys and Pure Metals*, Vol. 2, 9th ed., H. Baker (Managing Ed.), American Society for Metals, 1979. p. 41.)

Summary

- Ferrous alloys: steels and cast irons
- Non-ferrous alloys:
 - Al, Mg, Ti, Cu, Ni alloys; refractory metals; and noble metals.
- Metal fabrication techniques:
 - **forming**, **casting**, others (miscellaneous).
- Hardenability of metals
 - measure of ability of a steel to be heat treated.
 - increases with alloy content.
- Precipitation hardening
 - hardening, strengthening due to formation of precipitate particles.
 - Al, Mg alloys precipitation hardenable.

Strengthening of Metals



Ways of changing the strength of metals:

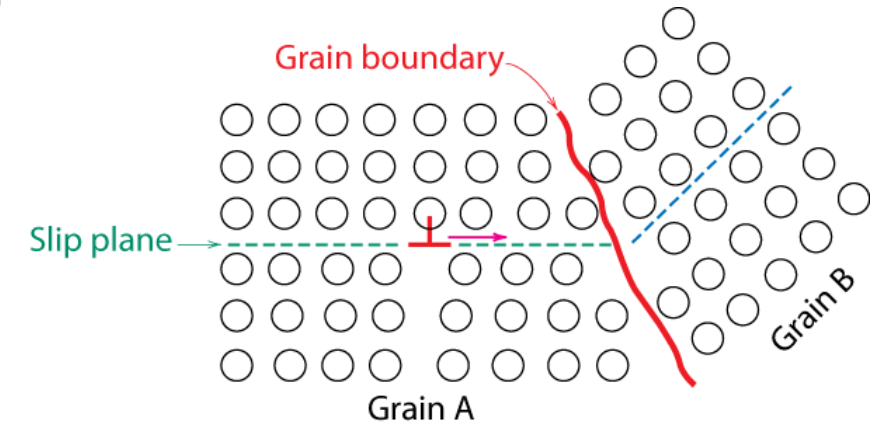
- Reduce Grain Size
- Form Solid Solutions (Solid-Solution Strengthening)
- Cold Work (Strain Hardening)

These can be changing when we process the material

Strategies for Strengthening:

1: Reduce Grain Size

- Grain boundaries acts as barriers to dislocation motion
- At boundary
 - Slip planes change directions
 - Discontinuity of slip planes
- Reduce grain size
 - Increase grain boundary area
 - More barriers to dislocation motion
 - Increase yield strength, tensile strength and hardness
- Dependence of σ_y on average grain diameter, d :



Adapted from Fig. 7.14, Callister & Rethwisch 8e.
(Fig. 7.14 is from *A Textbook of Materials Technology*, by Van Vlack, Pearson Education, Inc., Upper Saddle River, NJ.)

Hall-Petch Equation:

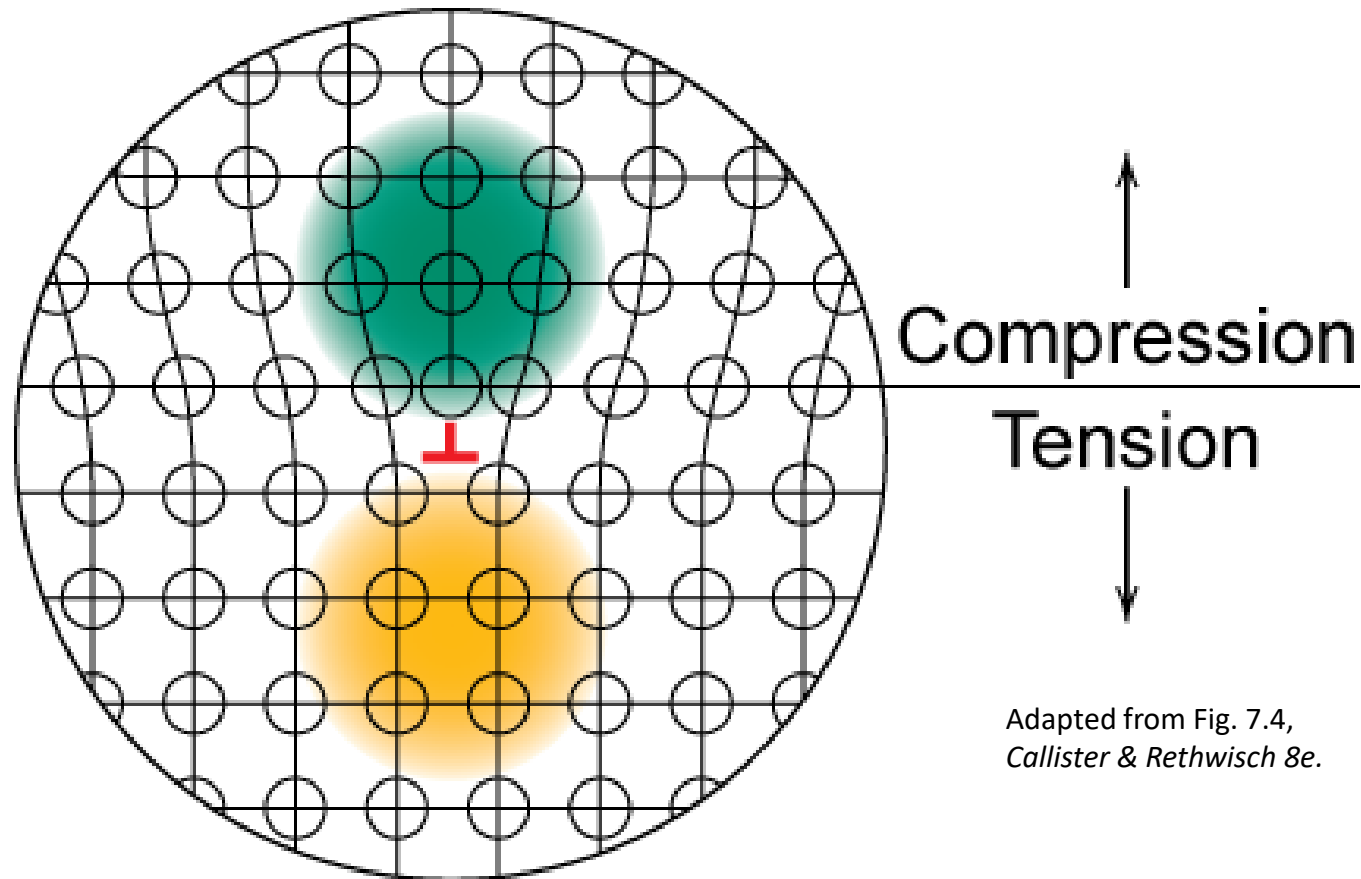
$$\sigma_{yield} = \sigma_o + k_y d^{-1/2}$$

σ_o , k_y = material constants

Strategies for Strengthening:

2: Form Solid Solution

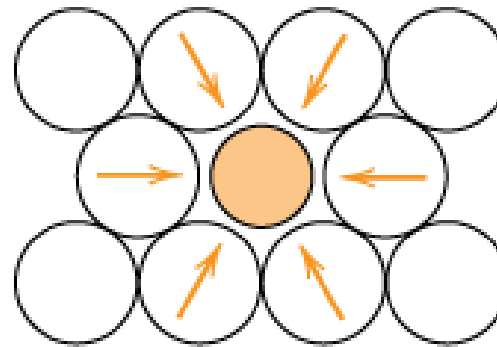
- Lattice strains around dislocations
 - Locations of tensile, compressive strains around an edge dislocation



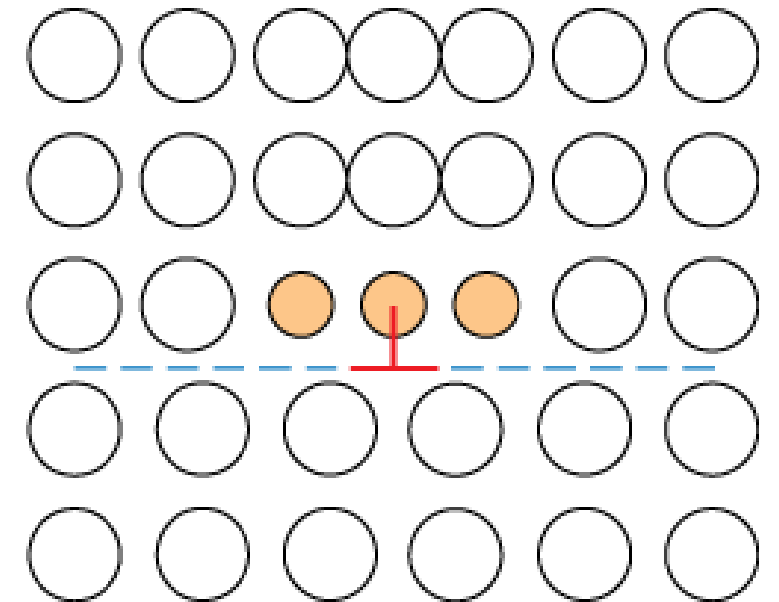
Adapted from Fig. 7.4,
Callister & Rethwisch 8e.

Strengthening by Solid Solution Alloying

- Lattice strain interactions with strains introduced by impurity atoms
- Small substitutional impurities introduce tensile strains
- When located above the slip line for edge dislocation:
 - partial cancellation of impurity (tensile) and dislocation (compressive) strains
 - higher shear stress required to cause dislocation motion



(a)

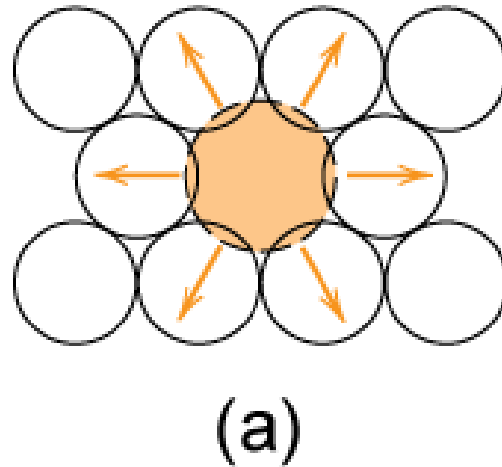


(b)

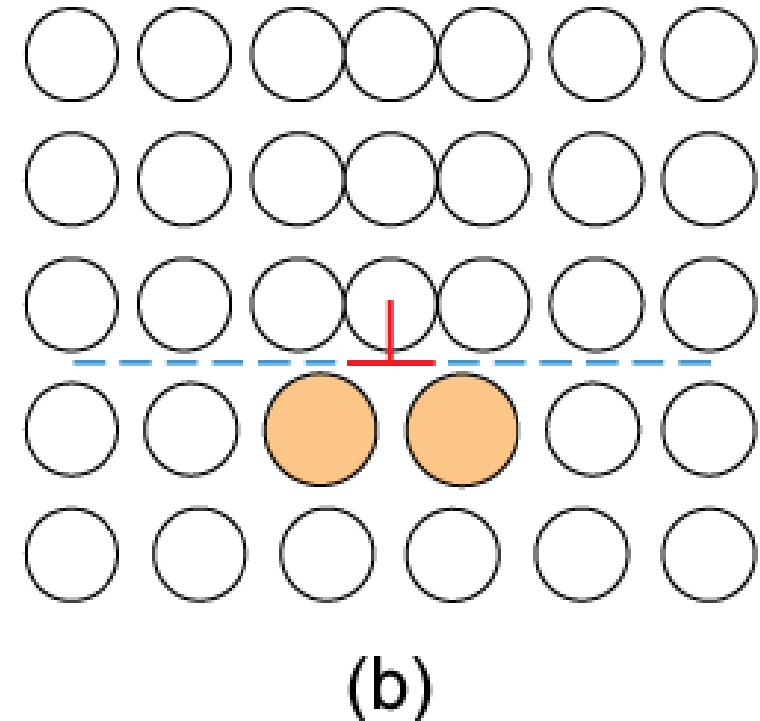
Adapted from Fig. 7.17, Callister
& Rethwisch 8e.

Strengthening by Solid Solution Alloying

- Large substitutional impurities introduce compressive strains
- When located below the slip line for edge dislocation:
 - partial cancellation of impurity (compressive) and dislocation (tensile) strains
 - higher shear stress required to cause dislocation motion



Adapted from Fig. 7.18,
Callister & Rethwisch 8e.



VMSE Solid Solution Strengthening Tutorial

[Main Menu](#)

Solid-Solution Strengthening

[Help](#)

This module allows you to observe the mechanism of solid solution strengthening by viewing animations, which are accompanied with voice-overs.



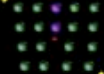
Pure



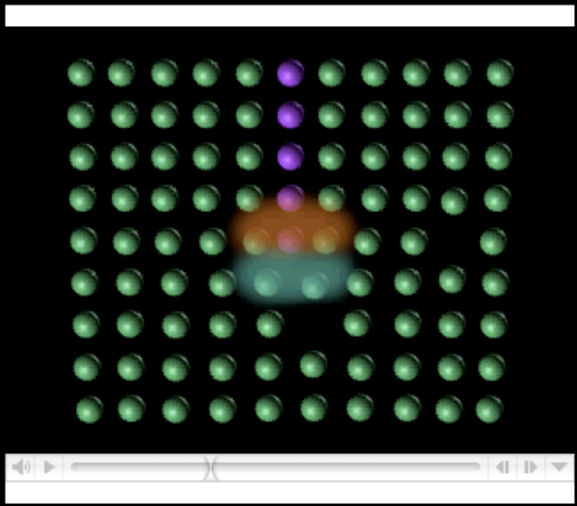
Larger



Smaller



Interstitial

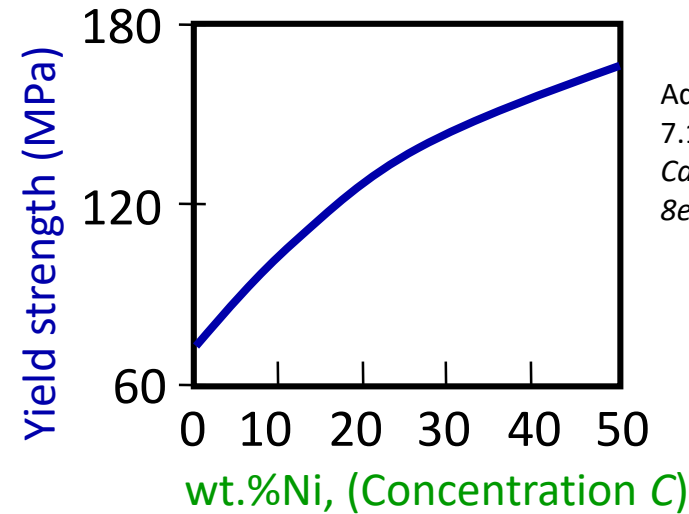
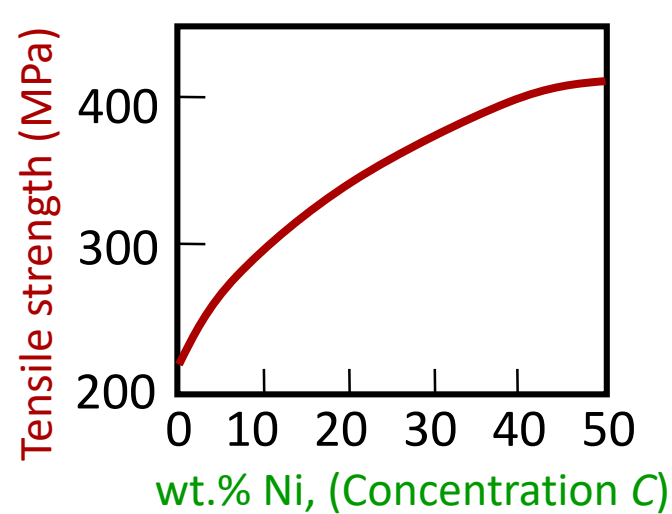


Please be patient. This may take a few minutes to load depending on your connection speed.

Ex: Solid Solution Strengthening in Copper



- Tensile strength & yield strength increase with wt% Ni.



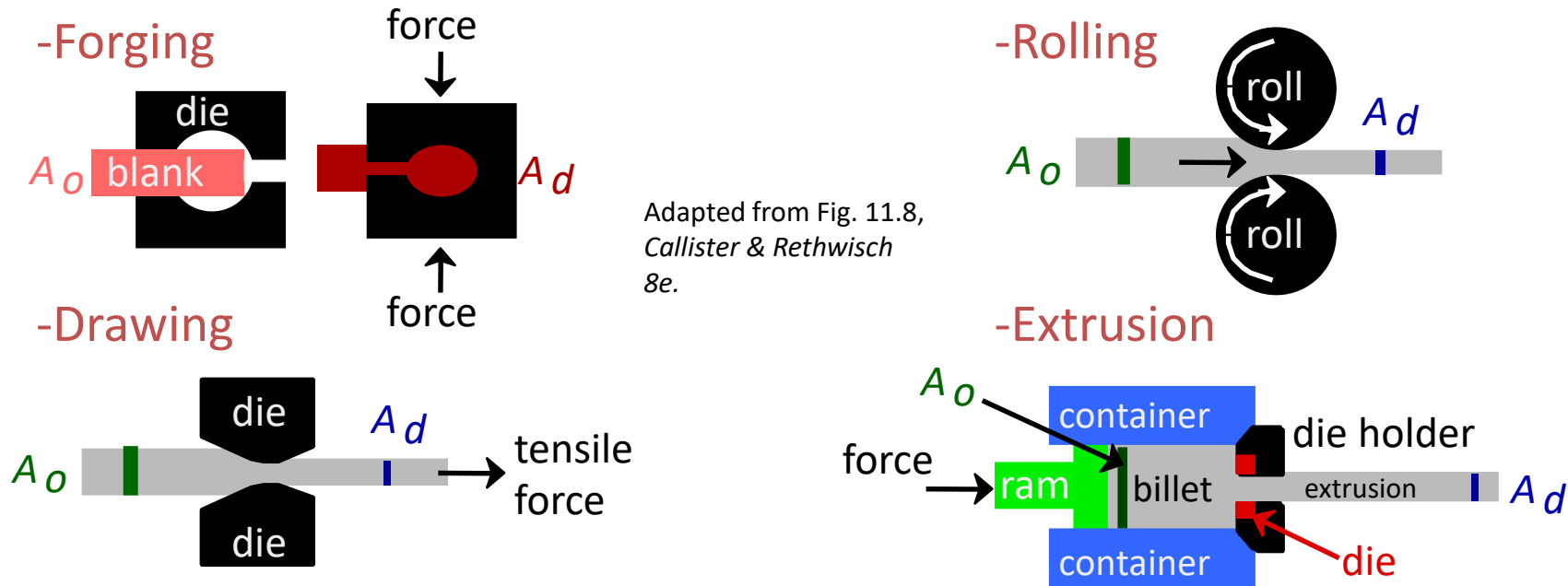
Adapted from Fig. 7.16(a) and (b), Callister & Rethwisch 8e.

- Empirical relation: $\sigma_y \sim C^{1/2}$
- Alloying increases s_y and TS .

Strategies for Strengthening:

3: Cold Work (Strain Hardening)

- Plastically deforming most metals at room temperature make them harder and stronger.
- Phenomenon referred to “Strain hardening (or cold working)”
- Deformation – often reduction in cross-sectional area



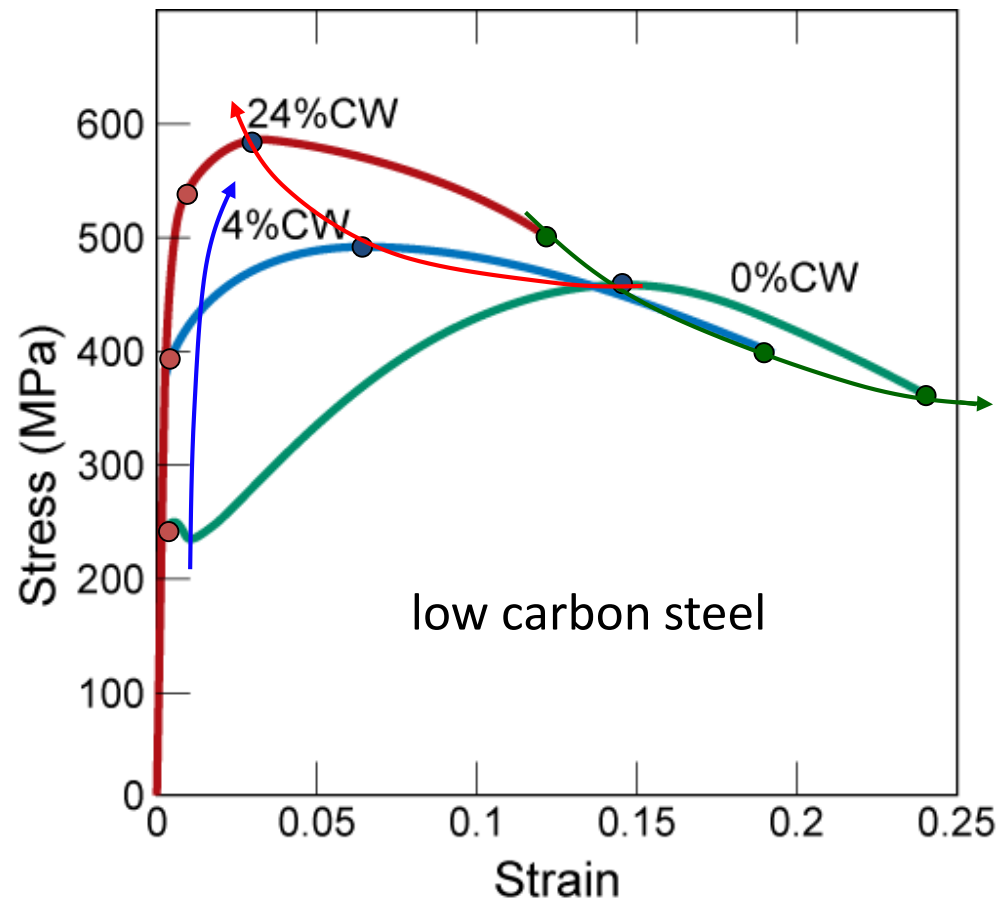
% Cold work (%CW) is calculated by

$$\%CW = \frac{A_o - A_d}{A_o} \times 100$$

Impact of Cold Work

As cold work is increased

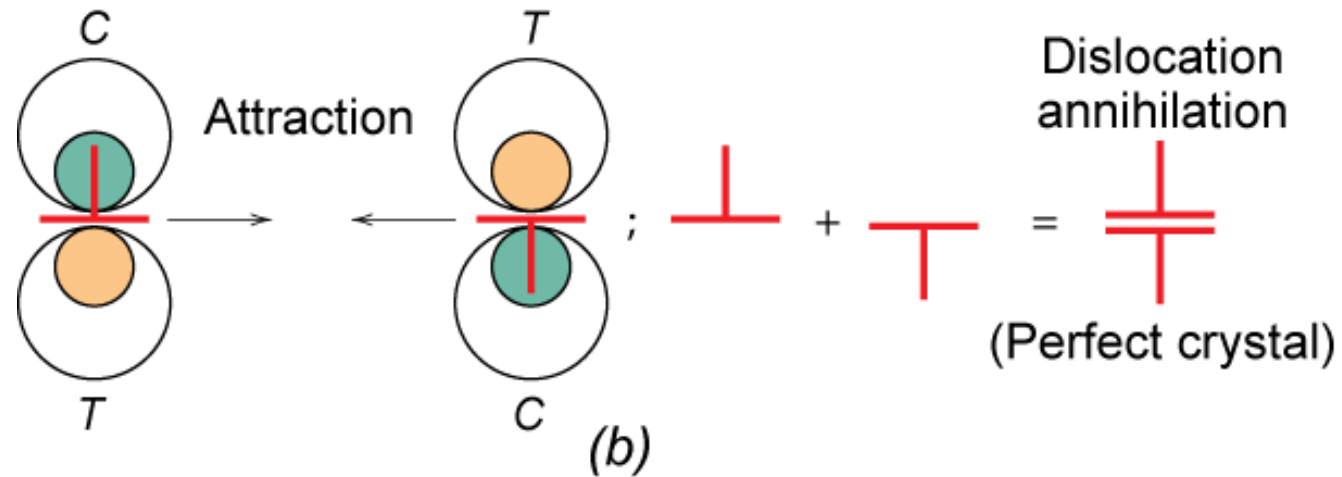
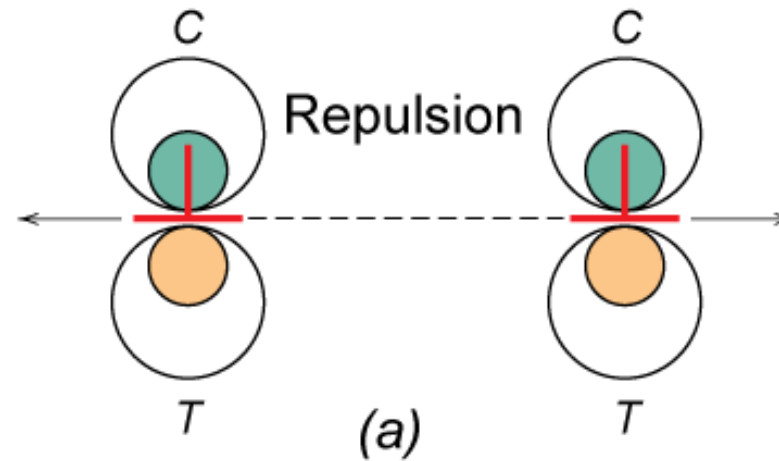
- Yield strength (s_y) increases.
- Tensile strength (TS) increases.
- Ductility ($\%EL$ or $\%AR$) decreases.



Adapted from Fig. 7.20,
Callister & Rethwisch 8e.

Lattice Strain Interactions Between Dislocations

Adapted from Fig. 7.5,
Callister & Rethwisch
8e.



Dislocation Density Increases During Cold Working

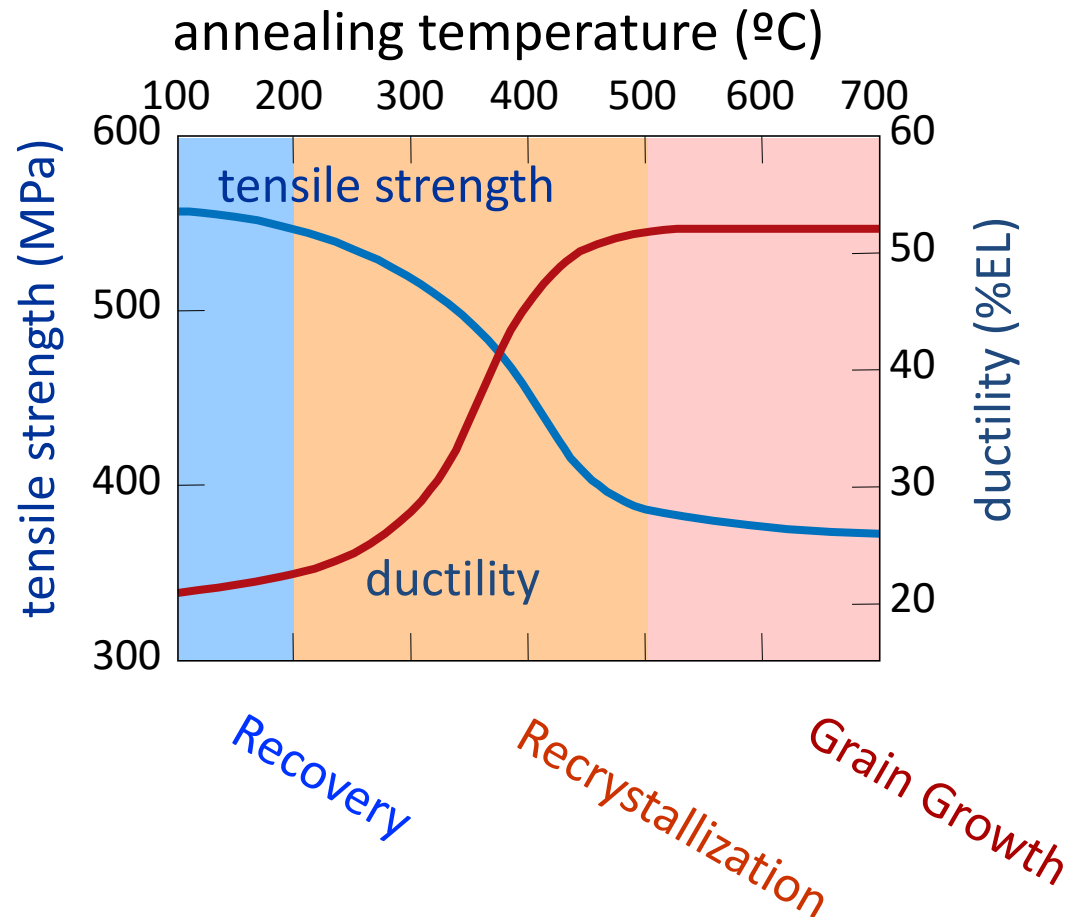
$$\text{Dislocation density} = \frac{\text{total dislocation length}}{\text{unit volume}}$$

- Dislocation density in underformed metal
→ $10^5 - 10^6 \text{ mm}^{-2}$
 - Dislocation density increases with increasing deformation
 - Dislocation density in derformed (cold-worked) metal
→ $10^9 - 10^{10} \text{ mm}^{-2}$
- Yield stress increases as r_d increases:

Effect of Heat Treating After Cold Working



- Heat treating cold worked metals brings changes in structure and properties
- As a result, effects of cold work are nullified
- This type of heat treatment is sometimes called “annealing”
- 1 hour treatment at T_{anneal} ... decreases TS and increases $\%EL$.



- Three Annealing stages:
 1. **Recovery** (100-200°C)
 2. **Recrystallization** (200-500°C)
 3. **Grain Growth** (>500°C)

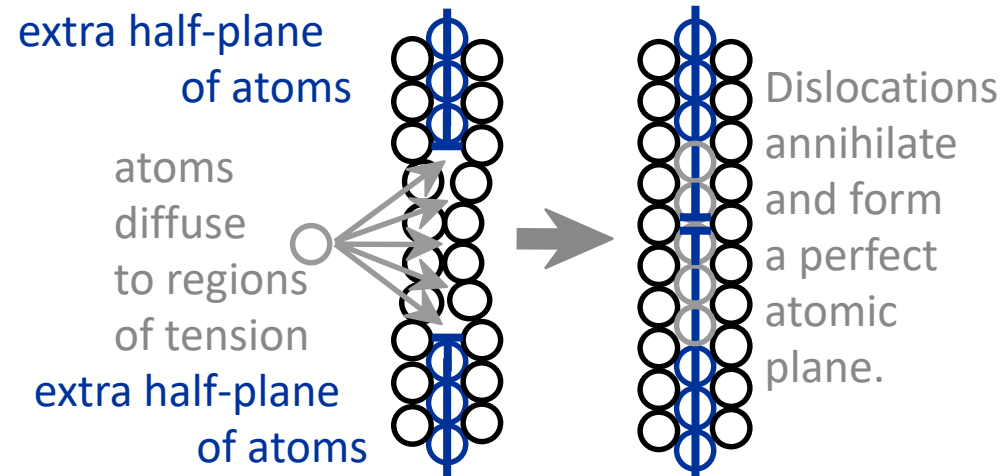
Adapted from Fig. 7.22, Callister & Rethwisch 8e.
(Fig. 7.22 is adapted from G. Sachs and K.R. van Horn, *Practical Metallurgy, Applied Metallurgy, and the Industrial Processing of Ferrous and Nonferrous Metals and Alloys*, American Society for Metals, 1940, p. 139.)

Three Stages During Heat Treatment:

1. Recovery

Reduction of dislocation density by annihilation.

- Scenario 1
Results from diffusion

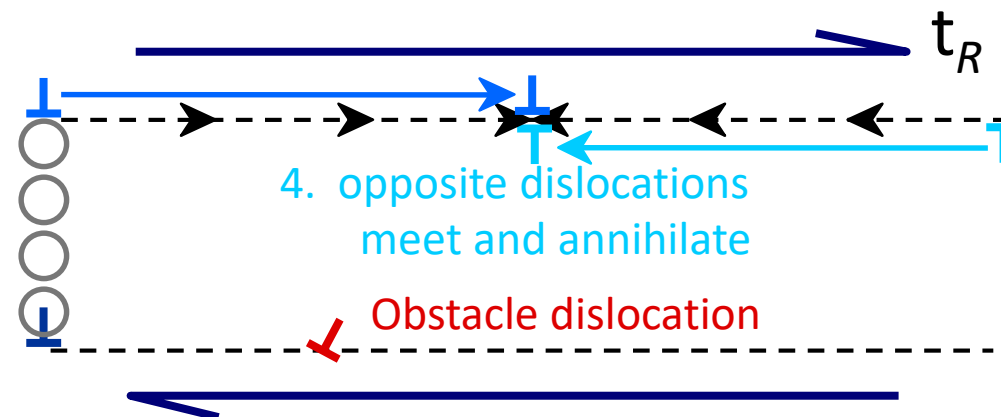


- Scenario 2

3. "Climbed" disl. can now move on new slip plane

2. grey atoms leave by vacancy diffusion allowing disl. to "climb"

1. dislocation blocked; can't move to the right



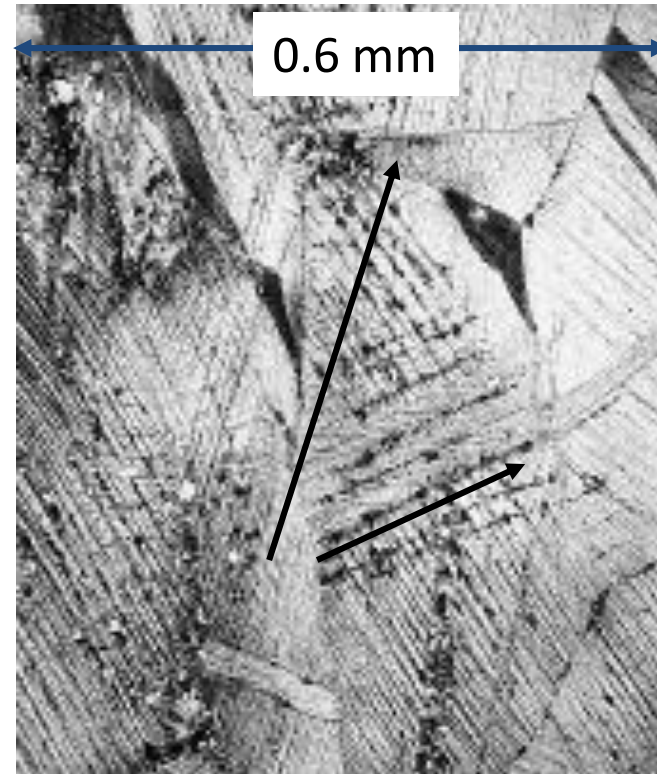
Three Stages During Heat Treatment:

2. Recrystallization

- New grains are formed that:
 - have low dislocation densities
 - are small in size
 - consume and replace parent cold-worked grains.



33% cold worked brass

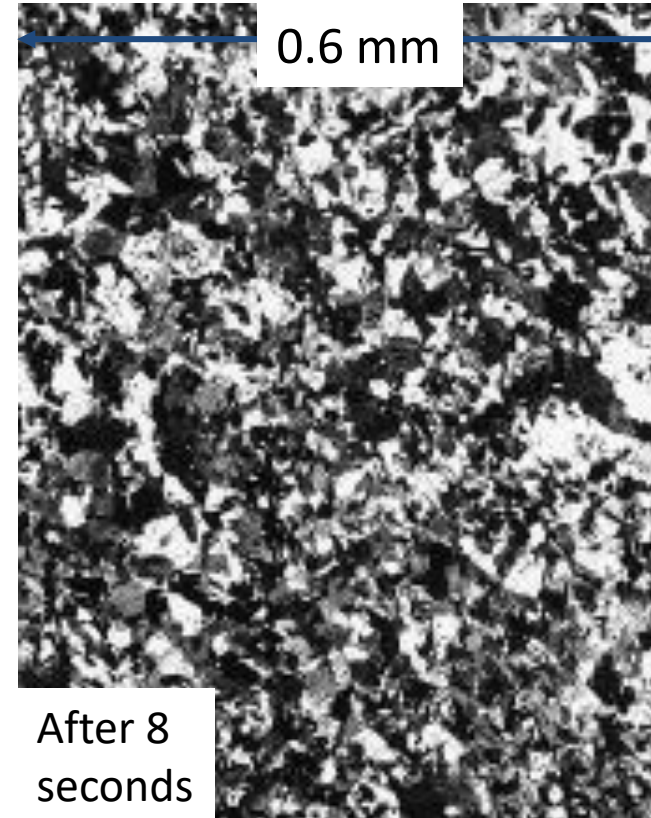
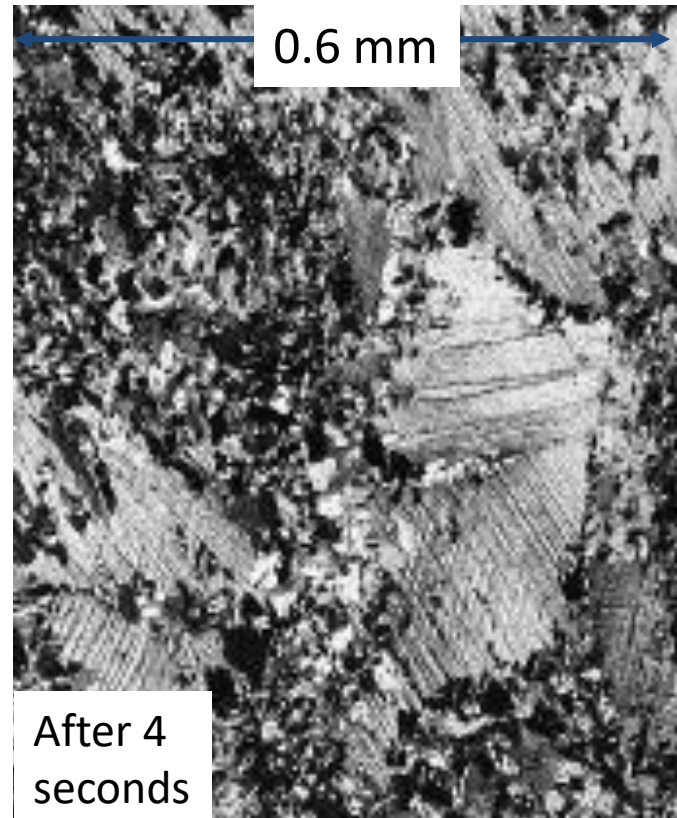


New crystals nucleate after 3 sec. at 580°C.

Adapted from Fig. 7.21(a),(b),
Callister & Rethwisch 8e. (Fig. 7.21(a),(b) are courtesy of J.E. Burke, General Electric Company.)

As Recrystallization Continues...

- All cold-worked grains are eventually consumed/replaced.



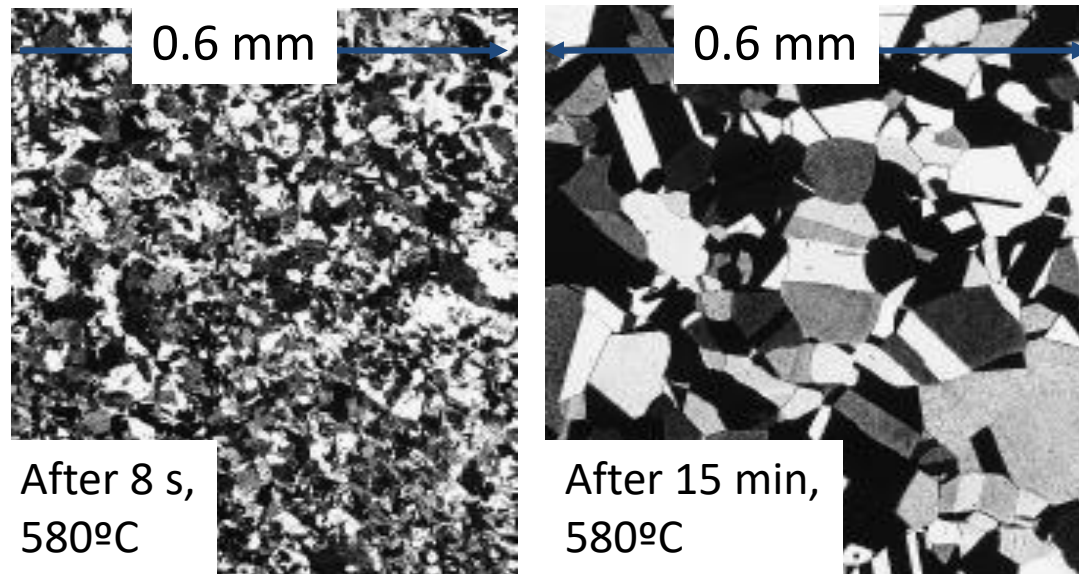
Adapted from Fig. 7.21(c),(d),
Callister & Rethwisch 8e. (Fig. 7.21(c),(d) are courtesy of J.E. Burke, General Electric Company.)

Three Stages During Heat Treatment:

3. Grain Growth



- At longer times, average grain size increases.
 - Small grains shrink (and ultimately disappear)
 - Large grains continue to grow



Adapted from Fig. 7.21(d),(e),
Callister & Rethwisch 8e. (Fig. 7.21(d),(e) are courtesy of J.E. Burke, General Electric Company.)

- Empirical Relation:

exponent typ. ~ 2

grain diam. at time t .

coefficient dependent on material and T .

elapsed time

$$d^n - d_o^n = Kt$$

Initial avg grain diam. b4 heat treatment.

Recrystallization Temperature

T_R = recrystallization temperature = temperature at which recrystallization just reaches completion in 1 h.

$$0.3T_m < T_R < 0.6T_m$$

For a specific metal/alloy, T_R depends on:

- %CW -- T_R decreases with increasing %CW
- Purity of metal -- T_R decreases with increasing purity

Cold Working vs. Hot Working

- Hot working $\rightarrow$ deformation above T_R
- Cold working $\rightarrow$ deformation below T_R

Grain Size Influences Properties

- Metals having small grains – relatively strong and tough at low temperatures
- Metals having large grains – good creep resistance at relatively high temperatures

Summary

- Dislocations are observed primarily in metals and alloys.
- Strength is increased by making dislocation motion difficult.
- Strength of metals may be increased by:
 - decreasing grain size
 - solid solution strengthening
 - cold working
- A cold-worked metal that is heat treated may experience recovery, recrystallization, and grain growth – its properties will be altered.